

FilaLoom Assembly Instructions

The following gives instructions for printing and assembling FilaLoom, please read these instructions carefully.

Printing

We recommend using PETG filament to print FilaLoom's components, as its higher elasticity results in smoother tool-changes compared to PLA. We used Polymaker PETG Blue.

All component geometries feature a flat side. When slicing the STL files, position the component on this side.

Slice components at standard settings, following the filament manufacturer's recommendations.

Assembly

The following give instructions for assembling FilaLoom's three components: Yarn Depositor, Print Head Cover, and Station. Each component consists of multiple parts. Attaching part together should be done using a cyanoacrylate adhesive (super glue).

Yarn Depositor

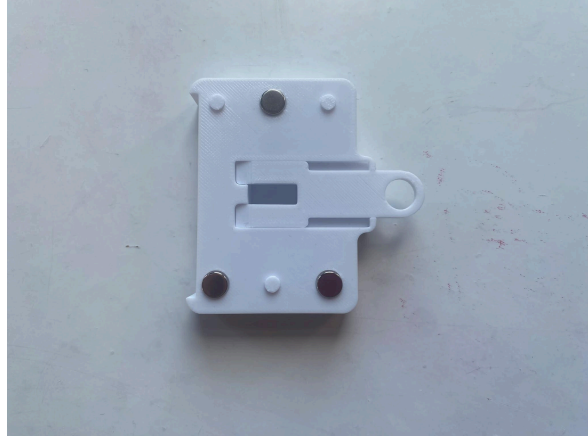
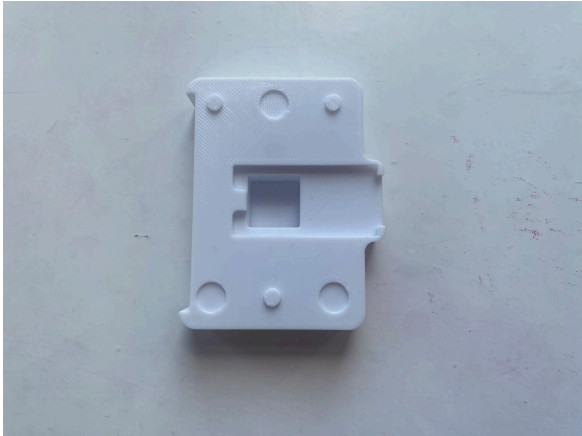
Printable parts:

Front Body
Back Body
Slider
Docking Bars
Needle Holder
Axle
Washer (2x)
Bolt
Spool

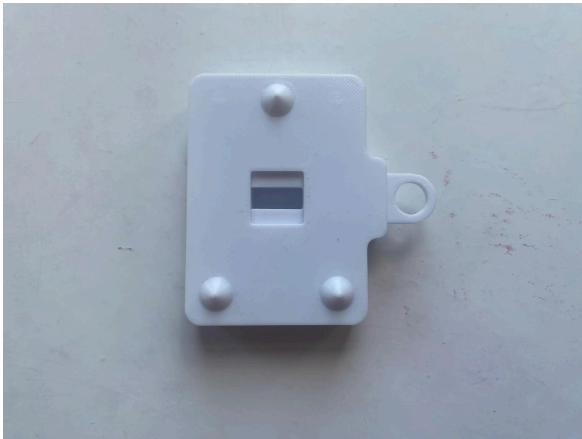
Non printable parts:

Compression spring; 10 mm diameter, 15 mm length, 0.5 mm wire thickness
Neodymium magnets (5x); cylinder shaped, 8mm diameter, 2mm height
Metal blunt tip dispensing needle; various sizes possible, we used 50mm long needles with 11G and 15G gauge

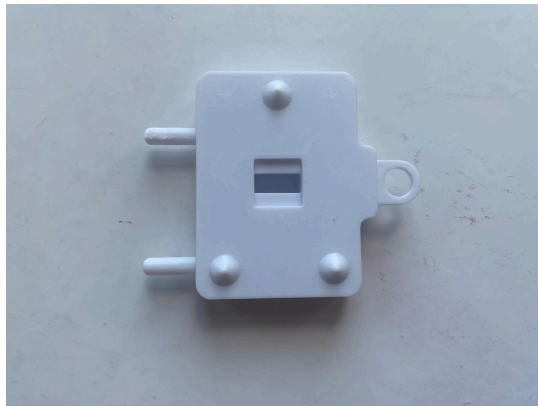
Assembly:



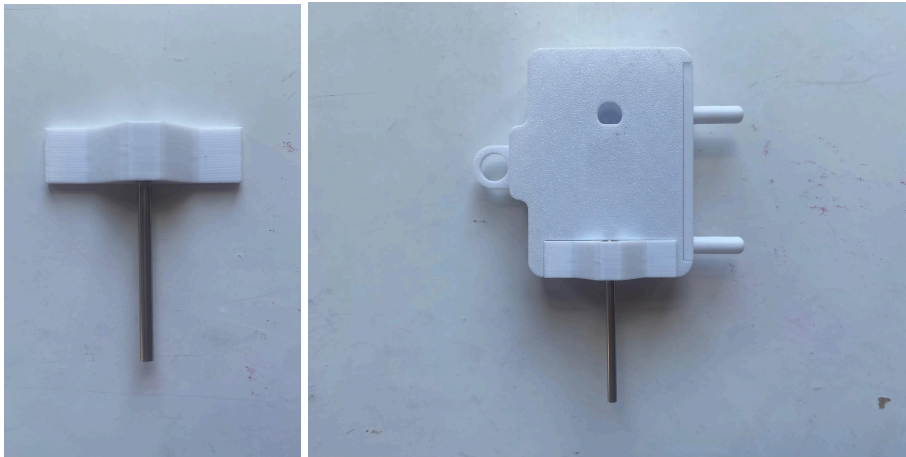
1) Lay the Front Body on a table with its flat side down, then place the Slider and three magnets inside of the corresponding holes in the Front Body. The Slider has a side which is slightly curved, make sure this side faces away from the Front Body.



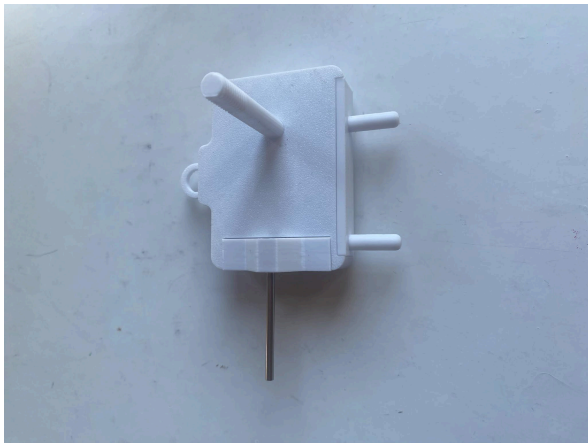
2) Glue the Back Body onto the Front Body, locking the slider and magnets inside of the yarn depositor body. Use their respective connection points to ensure good alignment. When glueing, make sure not to get glue on the Slider, as it should remain free to slide back-and-forth within the body.



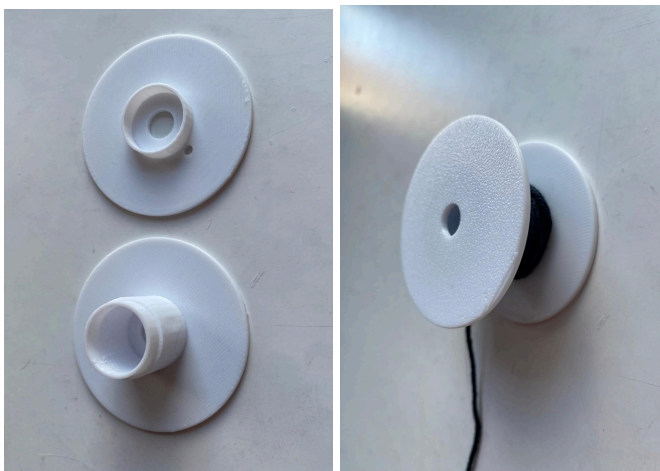
3) Place two magnets inside of the holes on the Docking Bars, and glue the Docking Bars into the corresponding cutouts on the side of the Front Body.



4) Push the Needle in the Needle Holder and glue the Needle Holder into the corresponding cutout on the flat side of the Front Body.



5) Place several drops of glue in the round hole at the flat side of the Front Body, and push Axle Inside



6) Attach the two parts of the Spool together and wind it with yarn.



7) Hang a Washer, the Spool, a second Washer and the Compression Spring onto the axle. Screw them in place using the Bolt

Print Head Cover

Printable parts:

Cover Front

Cover Back

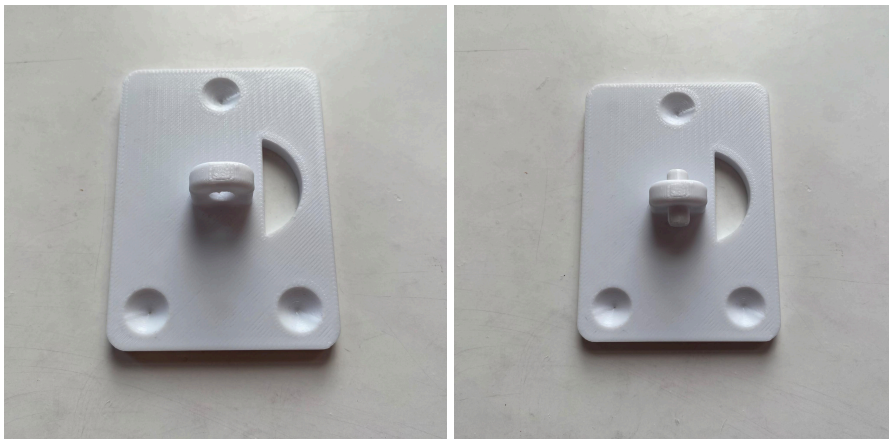
Positioner (3x)

Lock Cylinder

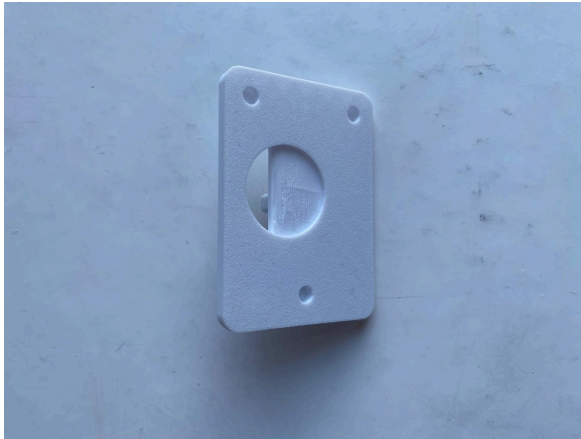
Non printable parts:

Neodymium magnets (3x); cylinder shaped, 8mm diameter, 2mm height

Assembly:



1) Push Lock Cylinder into the hole on the protrusion on Cover Front, ensure it is placed in the middle, making the protrusion have a +-shape when looking straight onto it.



2) Place the Positioners inside their corresponding holes on the Cover Front



3) Place three magnets on the flat side of the Cover Front. **Important:** Ensure the magnets pair with the three placed inside the yarn depositor body (they should attract each other). It is recommended to place the Cover Front onto the assembled yarn depositor when placing the magnets inside, the magnets will automatically snap into position.



4) Glue the Cover Back onto the Cover Front, Using the Positioners to align them properly.

5) Attach the assembled print head cover to the print head of the Bambu Lab A1 by clicking it in place.

Station

Printable parts:

Station

Anchor Hook

Non-printable parts:

Neodymium magnets (2x); cylinder shaped, 8mm diameter, 2mm height

M3 screws, 12mm long (2x)

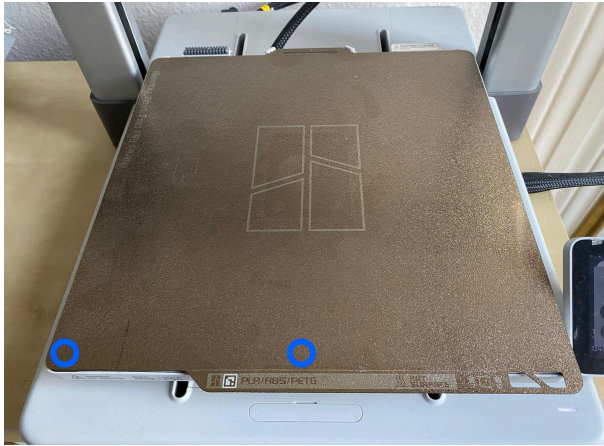
Assembly:



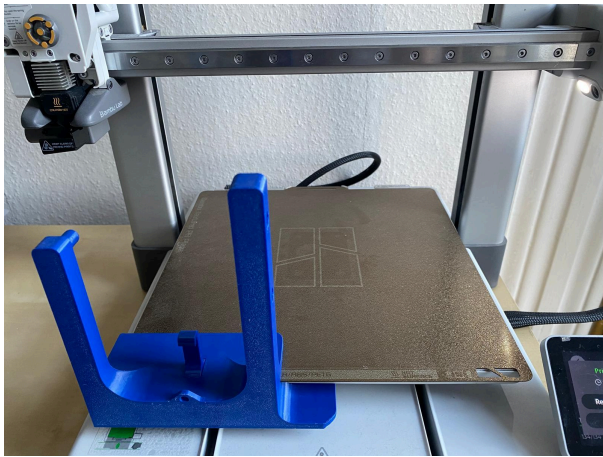
1) Glue magnets in corresponding holes on the large pillar of the station



2) Glue Anchor hook in the corresponding hole on the station



3) Unscrew the screws at the bottom of the print bed, on the front left and front middle



4) Slide station on print head and screw into the bottom using 12mm long M3 screws